

VJC 2022 H2 Biology Prelim Paper 3 Answer

Question 1

(a) (i) State, as precisely as you can, where Rubisco is found in the plants. [1]

- Stroma of chloroplast;

(ii) Outline the significance of Rubisco to the plants. [3]

- Carbon fixation in Calvin cycle;
- Catalyse addition of carbon dioxide to ribulose biphosphate to form unstable 6C compound, which is then split into 2 molecules of glycerate-3-phosphate;
- Allows carbon dioxide to be incorporated into Calvin cycle to produce glucose;

With reference to Fig. 1.1,

(iii) of all the caffeine-producing plants, state the genus of the least related one. [1]

- *Theobroma*;

(iv) deduce if the evolution of caffeine in plants shows divergent evolution. Give a reason for your answer. [2]

- No – convergent evolution;
- Ref. to plants evolving caffeine independently – not inherited from a caffeine-producing common ancestor;

(v) Justify the claim of the scientist that the phylogenetic tree is reliable and suggest an improvement to the reliability. [2]

Reliability

- Quantitative, not ambiguous, objective;
- Many points of comparison;
- Present in all plants;

Improvement

- Compare more than 1 homologous genes;
- Compare Rubisco gene sequence instead – silent mutations;

(b) With reference to Fig. 1.2 and 1.3,

(i) calculate the rate of increase in global consumption of coffee between 2015 and 2019. Show your working below. [1]

There are 4 years between 2015 to 2019.

From graph, 2015 consumption is 149 million 60kg bags. In 2019 it was 157.5 million 60kg bags.

Rate of increase in global consumption is $157.5 - 149 = 8.5$ million 60kg bags

In 4 years this will mean $8.5 / 4 = 2.125$ million 60kg bags / year

(ii) state the region that contributed the most to the increase in the global consumption of coffee. Give a reason for your answer. [2]

- Asia & Oceania;
- Largest increase + Q.V.;

(c) With reference to the information provided,

(i) explain how the structure of adenosine aids in its movement across the synapse. [2]

- Small and polar / hydrophilic;
- Allows it to diffuse quickly across the synapse;

(ii) explain the significance of cAMP in adenosine signalling. [3]

- Second messenger;
- Binds to and activates protein kinase A via conformation change;
- Trigger phosphorylation cascade of nociceptive pathway;

(iii) explain how caffeine functions as a cognitive enhancer. [3]

- Structural resemblance to (nitrogenous base of) adenosine;
- Ref. to competitive inhibition;
- Competes with adenosine to bind to adenosine receptor;
- Lowers number of bound receptor thus decreasing / preventing the increase of drowsiness;

(iv) suggest two reasons for the need of repeated consumption of coffee throughout the day in order to prevent feeling of drowsiness. [2]

- High concentration of adenosine as the day progresses – outcompete caffeine to bind to adenosine receptor;
- Removal / Metabolism of caffeine by the body;

(d) (i) Outline the symbiotic relationship between coffee berry borer and its gut bacteria. [3]

- Gut bacteria breaks down caffeine – allows coffee berry borer to consume coffee fruits and leaves as food source without adverse effect;
- Less competition for food with other herbivores which cannot consume coffee plants;
- Protection / Optimal environment for gut bacteria;
- Stable supply of caffeine as food source when coffee berry borer feeds on coffee fruits and leaves;

(ii) Explain how coffee berry borer may be classified as a distinct species. [2]

- Ref. to ecological species concept;
- Ref. to exploiting a new niche;

OR

- Ref. to biological species concept;
- Ref. to no interbreeding with other beetles to form fertile, viable offspring;

OR

- Ref. to morphological species concept;
- Ref. to longer gut;

Question 2

(a) (i) Outline how cancer cells can be 'stem cell-like'. [2]

- Cancer cells are undifferentiated cells with no tissue-specific structures or functions;
- Cancer cells have the ability to carry out cell division indefinitely / long-term;
- Cancer cells express the telomerase gene / have active telomerase;

(ii) Describe the differences that can be seen in cancer cells compared with normal germ line cells. [3]

feature	cancer cell	normal germ line cell
type of cell division	mitosis	mitosis and meiosis
exhibit contact inhibition	no	yes
need for growth factor	no	yes
cell cycle checkpoints	non-functional	functional
cell cycle arrest for DNA repair or to trigger apoptosis	continue to divide to accumulate mutations	cannot proceed with cell cycle when mutation / DNA damage is detected for repair or to trigger apoptosis
mutations in genes controlling cell cycle	yes	no
ability to induce angiogenesis and metastasis	yes	no

(b) (i) Suggest a reason why doctors remove the patient's stem cells from their blood or bone marrow before the cancer treatment begins. [1]

- To prevent cancer treatment from killing / causing mutations in the stem cells;

(ii) With reference to Fig 2.1, give a conclusion for the effectiveness of the two treatment methods for breast cancer in women. Justify your answer. [3]

- Both treatment methods show a decrease in effectiveness over time;
- Before 60 months, conventional-dose chemotherapy is more effective than high-dose chemotherapy plus stem-cell transplantation before 60 months – it takes around 60 months for probability of survival to decrease to 0.1 for conventional-dose chemotherapy, which is slower than the 55 months for high-dose chemotherapy plus stem-cell transplantation;
- After around 60 months, both methods are equally (in)effective;

Question 3

(a) With reference to Fig. 3.1, contrast the structure of SARS-CoV-2 with that of HIV. [2]

feature	SARS-CoV-2	HIV
type of glycoprotein on viral envelope	spike protein (S)	gp120 and gp41
no. of single strands RNA	one	two copies
presence of capsid coat	absent	present
presence of viral enzymes	no viral enzymes	contains reverse transcriptase and integrase

(b) Describe three ways in which the structure of antibodies contributes to their functions. [3]

- Specific antigen binding sites that are complementary in shape to specific antigens – allow binding to antigens;
- 2 antigen binding sites – allow binding to 2 antigens;
- Fc region – binds to specific Fc receptor on specific immune cells for effector functions;
- Disulfide bonds – maintain specific 3D conformation;

(c) With reference to Fig. 3.2, explain how variability at the DNA level result in variability in the antigen-binding sites of antibodies. [3]

- Ref. to somatic recombination;
- Results in random combinations of VJ gene segments in the light-chain loci and VDJ gene segments in the heavy-chain locus;
- Ref. to expression of the VJ gene segment in either the λ light-chain locus on chromosome 22 or κ light-chain locus on chromosome 2;
- Expression of one particular VJ gene segment and one particular VDJ gene segment results in one particular light chain variable domain and one particular heavy chain variable domain respectively, which together form an antigen binding site of a particular shape and specificity;

(d) Explain how the vaccine triggers the production of antibodies and how the antibodies confer protection against the virus. [5]

How vaccine triggers production of antibodies (4 max.)

- Ref. to antigen presentation by antigen presenting cells;
- Specific helper T cell activated to release cytokines;
- Specific naïve B cell activated, aided by cytokines released by helper T cell;
- Activated B cells undergo clonal expansion and differentiation into plasma cells that release antibodies specific to the spike proteins;

How antibodies confer protection against the virus

- Antibodies bind to the spike proteins on SARS-CoV-2 to block attachment to specific receptor on host cells, thus preventing infection / facilitate phagocytosis of by macrophages;

(e) Suggest why scientists have not been able to create an effective vaccine against HIV. [1]

- There is high mutation rate in the HIV RNA genome due to the high error rate and lack of proofreading in HIV reverse transcriptase;

Question 4

(a) Epigenetics is the study of changes in organisms that are caused by modifications of gene expression rather than alteration of the genetic code. Describe how such modifications can be achieved. [15]

Chromatin remodelling

1. The packaging of **DNA with histones into nucleosomes/chromatin** can allow for **chromatin remodeling** which can modify gene expression
2. **Histones acetylation** at the **free lysine residues at the N terminus**
3. by **histone acetylase** (accept if histones deacetylation is mentioned instead)
4. result in **decreased net positive charge of histones**
5. **lowering the affinity of histones to negatively charged DNA**
6. to form **euchromatin/become less condensed**
7. This allows for **greater accessibility of general transcription factors and RNA polymerase** at the promoter to **increase transcription** (and specific transcription factors at control elements, to be covered in the second portion of answer)
8. **Histone deacetylation** will result in **chromatin becoming more condense and reduce transcription**
9. **DNA methylation** adds methyl groups to **DNA sequences rich in cytosine / CpG-rich sites**
10. catalysed by **DNA methyltransferase**
11. Many transcription factors bind to DNA at CpG-rich sites, thus methylation of these sites **interferes with transcription factors binding**, resulting in the **suppression of the initiation of transcription**.
12. It also induces **histones deacetylation** as **histones deacetylase** recognizes and **bind to the methylated DNA**

Increase accessibility to specific transcription factors

13. Each gene has a **specific/unique combinations** of **control elements** like **enhancers or silencers**.
14. The presence of **appropriate combination of activators or repressors proteins** in a particular cell types at a precise time can determine the expression of the genes
15. **Activators** can bind to the **enhancer** to **upregulate** the RNA polymerase activity/transcription
16. **Repressors** can bind to the **silencer** to **downregulate** the RNA polymerase activity/transcription
17. the binding of specific transcription factors to the control elements will trigger a **DNA looping mechanism aided by DNA bending proteins**
18. to bring the **specific transcription factors close** to the promoter to **interact with the transcription initiation complex**.

QWC – cover 2 aspects of chromatin remodelling (DNA methylation and histones acetylation)

(b) Using relevant examples, discuss the significance of regulating gene expression. [10]

Introduction:

1. Regulation of gene expression is the process which controls the type, timing (temporal), location (spatial) and amount of genes expressed in a cell;

Prevent wastage of energy/resources

2. Allow prokaryotes to **synthesise only enzymes/proteins that are required** and thus **prevent wastage of energy and resources**
3. (prokaryotes) Using trp operon as an example, in the presence of excess tryptophan, trp operon is switched off;
4. As tryptophan is in excess, no need to express enzymes involved in tryptophan synthesis;
5. Relevant detail for explanation e.g. trp acts as co-repressor, activates trp repressor, binds to operator, block RNA polymerase from binding
OR
6. Or lac operon is only expressed **in the absence of glucose and presence of lactose**;

7. As bacteria prefer to use glucose as respiratory substrate, there is no need to express enzymes required for lactose metabolism when glucose is present;
8. Relevant detail to explain catabolite repression e.g. in the presence of glucose, cAMP levels are low → no active cAMP-CAP complex → lac operon is switched off

Allows for cell differentiation

9. (Almost) all cells (except gametes) of a multicellular organism have the same set of genes but different cells switch on different genes to express only the proteins required for their specialized functions
10. For example beta cells of islet of Langerhans express genes for insulin and switch off genes for glucagon / vice versa for alpha cells
OR neuron cells express genes for neurotransmitters but beta cells of islet of Langerhans switch those genes off
OR as blood stem cells differentiate into red blood cells, genes expressing haemoglobin are switched on
OR as activated B cell differentiate into plasma cells, genes expressing antibodies are upregulated (Accept another valid examples)

Response to environment / maintenance of homeostasis

11. Cells divide in response to growth factors
12. When growth factors bind to their receptors, cell signalling pathway(s) is set off, which activate certain transcription factors
13. Which activate the expression of genes controlling/promoting cell division
OR
14. Antigen binding activates T helper cells and upregulate expression of cytokines
15. Activated B cell in response to cytokines differentiate into plasma cells upregulating expression of antibodies to be secreted
OR
16. Liver/muscle cells' response to insulin/glucagon (high/low blood glucose concentration)
17. When named hormone bind to their receptors, cell signalling pathway(s) is set off, which activate phosphorylation cascade / phosphorylation of named enzymes of the signal transduction pathway
18. Which activate named enzymes which catalyse specific reactions (cellular response) to restore normal blood glucose concentration

Development

19. (multicellular organisms) express the genes only at required times of their developmental process
20. Example globin genes are expressed at different time and in different amounts in human development, i.e. foetal Hb is different from adult Hb due to expression different globin genes (or any other relevant examples)
21. Stem cells express telomerase to allow indefinite self-renewal while differentiated cells silenced telomerase gene
22. Honey bee queen and worker bees have the same set of genes but differences in the DNA methylation profile of the queen and worker larvae result in different genes being activated and hence different phenotypes being produced
23. Queen larvae are fed with royal jelly which results in increased expression of genes in ovary development
24. Royal jelly activates pathways associated with the catabolism of proteins, carbohydrates and fats → increased growth rates seen in queen larvae relative to worker larvae

Regulation of cell cycle to prevent cancer

25. Expression of cell cycle checkpoint proteins is regulated to ensure cell divides in a controlled manner without accumulating mutations
26. Regulation of Ras protein activity by GTP – example of post-translational modification / expression of Ras gene is regulated to prevent overexpression
27. Prevent overstimulation of cell division – prevents uncontrolled division

28. Regulation of p53 expression – only expressed in response to appropriate signals e.g. detection of DNA damage

QWC – cover at least 2 categories with relevant examples;

Question 5

(a) Respiration produces ATP. Describe the role of ATP in living organisms. [15]

Introduction

1. Brief outline of ATP structure – **adenine nitrogenous base + ribose sugar + 3 phosphate groups**
2. **Small, water-soluble energy carrier** molecule
3. **Hydrolysis of ATP** / cleavage of phosphate group **releases energy** to drive **cellular / metabolic / endergonic (energy-requiring) reactions**

Protein synthesis

4. **Amino acid activation** / attachment of amino acid to tRNA

Transport across membrane

5. ATP is required for **active transport** to transport substances **against concentration gradient**
6. E.g. ATP binds to and phosphorylates sodium-potassium pump, causing it to change conformation so that sodium leaves while potassium binds (or any other valid example)
7. ATP is required for **bulk transport** like **exocytosis or endocytosis**
8. E.g. ATP required for movement of (any named) vesicles / motor proteins that drag the vesicles

RNA monomer

9. **Monomer of ribonucleic acid (RNA)**
10. The **2 terminal phosphate groups of each ATP are hydrolysed** and the nucleoside monophosphate is **incorporated into the RNA strand** / forms phosphodiester with another nucleotide

Cell signaling

11. Involved in **phosphorylation cascade** of a cell signaling pathway,
12. ATP is used to **activate proteins by phosphorylating them** / each **kinase catalyzes the addition of phosphate groups from ATP to a protein**
13. **cAMP, a second messenger** molecule involved in relaying signals, is synthesized **from ATP, catalysed by adenylyl cyclase**

Respiration

14. ATP used to **phosphorylate glucose in glycolysis**
15. **Activate / raise energy level / increase reactivity of glucose** at the start of
16. Makes sugar molecule **negatively-charged and unable to diffuse** out of the cell
17. **Maintains a steep concentration gradient** for glucose to enter the cell

Photosynthesis

- 18a. ATP is used in Calvin cycle to reduce **3-phosphoglycerate to triose phosphate**
- 18b. It is also used to **regenerate ribulose biphosphate**

Prokaryotes

19. In prokaryotes, **ATP can be converted to cAMP** by adenylyl cyclase to bind to **catabolite activator protein (CAP)**
20. **cAMP-CAP complex** can bind to **CAP binding site** to enhance RNA polymerase binding to promoter to **upregulate expression of lac operon**.

QWC (1 mark): Cover at least 2 areas for the role of ATP

(b) Using relevant examples, discuss the significance of complementarity in the production of ATP. [10]

Introduction

- 1 Complementarity refers to the interaction between two molecules that allows them to **bind** together / are **complementary in terms of shape/size/charge/orientation**.

Significance in substrate level phosphorylation

- 1 ATP is produced via **substrate-level phosphorylation** in **glycolysis and Krebs cycle**
- 2 which is the **direct enzymatic transfer of a phosphate group from a substrate to ADP** to form **ATP**
- 3 enzyme **active site** must be **complementary in shape, (size, charge and orientation) to substrate**
- 4 Named enzyme, substrate and products of any one SLP reaction

Significance in oxidative phosphorylation

- 5 ATP also produced via **oxidative phosphorylation / chemiosmosis**
- 6 electron transport chain proteins **has specific binding site** – complementary in **shape to reduced NAD and FAD** to receive electrons carried on them
- 7 Some **ETC complexes come into contact with each other** as they pass electrons – idea of complementarity
- 8 Also have **specific channels for H ions/protons** (idea than channel must be complementary in shape and size to H ions)
- 9 to **pump H ions /protons across the membrane / ref to active transport of H ions/protons to maintain proton gradient**
- 10 ATP **synthase catalyse formation of ATP from ADP and Pi**
- 11 **ATP synthase has specific active site** – complementary in **shape to its substrates ADP and Pi**
- 12 And **specific channel for H ion/protons to pass through via facilitated diffusion**
- 13 **Proton flow** provides energy required for **ATP synthesis**

Significance in anaerobic respiration

- 14 In anaerobic respiration, **enzyme (lactate dehydrogenase/alcohol dehydrogenase)** has **specific active sites to substrate to regenerate NAD for continual ATP production**

QWC (1 mark): At least two correct examples mentioned.